

What is special about a programmable power supply

All programmable power supplies must be accurate, linear, reliable and programmable. By programmable we mean that, depending on T&M needs, voltage and current should be controlled as required by the testing that needs to be done. Critical parameters on which we work are response time, linearity and accuracy. Some applications of programmable switching-mode power supplies are in transformers, motors, switchgear testing and solar emulation.

—Vijay Kambe, partner, KB Electronics

true 12-bit oscilloscopes in the market with software enhancement to 15 bits, providing the necessary accuracies that may be required for precision design. "We also introduced 8-channel scopes to address the 3-phase power design requirement and with motor-drive-analysis software, making it a versatile design, testing and analysis tool for any motor development. It provides flexibility of time-domain measurement and also specifics to power design at any stage of the circuit, whether it is analogue, digital or protocol signals, thus giving complete information of both electrical and mechanical measurements," says Rajendran Thanu, speaking on behalf of Teledyne Lecroy. This instrument targets design engineers and manufacturers working on power supplies, including those working on 3-phase power supply designs and motor controllers or drivers.

Keithley introduced two high-voltage power supplies optimised for high-voltage devices and materials-testing and high-energy physics and materials-science research. Model 2290-5 5kV power supply and model 2290-10 10kV power supply are well-suited for high-voltage breakdown testing of power semiconductor components. This includes devices made of current- and next-generation wide-bandgap materials like silicon carbide (SiC) and gallium nitride (GaN), designed for use in green, energy-efficient power generation and transmission systems and hybrid and all-electric vehicles.

Intuitive user interfaces could be the next big thing

It was about a decade ago that user interface design began to define the success or failure of consumer products. It now seems that test instrument vendors are testing out if there could be a similar effect in the engi-

neering tools and equipment space.

One such change is in battery simulators. In contrast to usual battery simulators targeting manufacturing applications, Keithley 2281S incorporates a thin-film transistor display that shows voltage, current, amp-hour readings, source settings and many additional settings. An icon based main menu provides access to source setup, measurement setup, display formats, trigger options and system settings.

A really interesting one is the new GUI on N89xxAPV that allows you to drag-and-drop a cloud or rainy environment, or change the time of rain. When manufacturers make these solar products, they create it based on the region they want to serve. So a US-focused product might be tuned to work without a lot of sun, but an Indian version might get a lot of sun.

The simulator allows these to test micro-inverters based on region by easily changing weather conditions, and the GUI makes it easier than a textual format. Coding using C or C# or other scripts are time consuming and difficult to get right. Previously, some original equipment manufacturers were even depending on pre-configured tests provided from other manufacturers and as a result were limited to those tests only.

Is it time to take a step back

While most specification-level feature advances are always welcomed by engineers, some online forums have often had this discussion of whether test equipment really needs touchscreens, remote control and other features inspired from consumer electronics. In fact, a recent survey done at *EDN Network* showed that more than half of the engineers who voted opted for knobs and buttons rather than touchscreens and tablets. Interesting, isn't it? ●

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